

PROJECT SHOWCASE

Sales Analytics End-to-End Project

ApexPlanet Data Analytics Internship

Capstone Portfolio — Task 5

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Project Overview

The **Sales Analytics End-to-End Project** is the capstone project completed during the **ApexPlanet Data Analytics Internship**. It demonstrates the complete lifecycle of a real-world data analytics project, transforming raw sales data into meaningful business insights through data preparation, exploratory analysis, business intelligence, interactive dashboards, statistical validation, and executive storytelling.

The project emphasizes practical problem-solving and business decision support using industry-standard tools and technologies.

Project Objectives

- ▶ Clean and prepare raw sales data for analysis.
- ▶ Perform comprehensive Exploratory Data Analysis (EDA).
- ▶ Generate business intelligence using SQL.
- ▶ Develop interactive Power BI dashboards.
- ▶ Validate business insights using statistical techniques.
- ▶ Communicate findings through executive storytelling.
- ▶ Build a portfolio-ready analytics project.

End-to-End Analytics Workflow



8. Business Storytelling



9. Professional Presentation & Portfolio

Internship Tasks

Task 1 — Data Immersion & Wrangling

Objectives

- Data Profiling
- Data Cleaning
- Missing Value Treatment
- Feature Engineering
- Outlier Detection
- Data Validation

Deliverables

- Data Profiling Notebook
- Data Cleaning Notebook
- Data Dictionary
- Data Quality Report
- Cleaned Dataset

Task 2 — EDA & Business Intelligence

Objectives

- Exploratory Data Analysis
- KPI Development
- Revenue Analysis
- Customer Analysis
- Product Performance
- SQL Business Intelligence

Deliverables

- EDA Notebook
- BI Notebook
- SQL Queries
- Business Insights Report
- Executive KPI Summary

Task 3 — Deep-Dive & Dashboarding

Objectives

- Dashboard Development
- Executive KPI Monitoring
- Customer Analytics
- Revenue Analysis
- Dashboard Documentation

Deliverables

- Power BI Dashboard
- Dashboard PDF
- Dashboard Screenshots
- Deep-Dive Report

Task 4 — Storytelling & Statistical Validation

Objectives

- Statistical Validation
- Business Storytelling
- Executive Reporting
- Business Recommendations

Deliverables

- Validation Notebook
- Storytelling Notebook
- Statistical Report
- Storytelling Report
- Presentation

Executive KPIs

₹139.40M	992	947
TOTAL REVENUE	TOTAL ORDERS	TOTAL CUSTOMERS
5,435	₹140.52K	₹147.20K
QUANTITY SOLD	AVG ORDER VALUE	REVENUE / CUSTOMER

Key Business Insights

Revenue Performance

- ▶ Generated total revenue of **₹139.40 Million**.
- ▶ Average Order Value reached **₹140.52K**.
- ▶ Revenue per Customer averaged **₹147.20K**.

Product Performance

- ▶ Electronics generated the highest revenue by category.
- ▶ Laptop emerged as the top revenue-generating product.
- ▶ Mobile recorded the highest quantity sold.

Customer Analysis

- ▶ Adult customers contributed the highest revenue.
- ▶ Revenue distribution remained balanced across genders.
- ▶ Spending patterns were statistically consistent across age groups.

Geographic Analysis — Top Cities by Revenue

1. Patna
2. Kolkata
3. Bengaluru
4. Mumbai
5. Hyderabad

Dashboard Highlights

The project includes two interactive Power BI dashboards providing executive-level visibility into sales performance and customer behavior.

Executive Overview	Customer Deep Dive
▶ Executive KPI Cards ▶ Revenue Trend ▶ Revenue by Product ▶ Revenue by Category ▶ Revenue by City ▶ Monthly Revenue Trend	▶ Revenue by Gender ▶ Revenue by Age Group ▶ Top Customers ▶ Orders by Category ▶ Average Order Value ▶ Interactive Filters

Statistical Validation

Business findings were rigorously validated using statistical techniques to ensure data-driven confidence in all recommendations.

Techniques Applied

- ▶ Descriptive Statistics
- ▶ Outlier Detection (IQR Method)
- ▶ Pearson Correlation Analysis
- ▶ Independent Sample T-Test (Welch's)
- ▶ One-Way ANOVA
- ▶ 95% Confidence Interval Analysis

Key Statistical Findings

- ▶ Strong positive correlation between Quantity Sold and Total Sales ($r = 0.65$).
- ▶ No statistically significant difference in spending between Male and Female customers ($p = 0.495$).
- ▶ Customer spending remained consistent across Age Groups (ANOVA $p = 0.973$).
- ▶ Statistical validation strengthened confidence in all business recommendations.

Technology Stack

Category	Technologies
Programming	Python, Pandas, NumPy
Database	SQL (MySQL Syntax)
Statistics	SciPy — Pearson Correlation, T-Test, ANOVA, CI Analysis
Visualization	Matplotlib, Seaborn, Power BI
Dev Tools	Jupyter Notebook, Git, GitHub, Microsoft PowerPoint

Skills Demonstrated

Technical Skills

Python	SQL	Pandas	NumPy
SciPy	Power BI	Git & GitHub	Jupyter

Analytics Skills

Data Cleaning	Feature Engineering	EDA	Business Intelligence
Dashboard Dev	Statistical Analysis	KPI Development	Data Storytelling

Business Skills

Executive Reporting	Business Analysis	Decision Support	Strategic Recs
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Project Achievements

- ★ Successfully completed all four internship tasks plus capstone portfolio.
- ★ Built a complete end-to-end Sales Analytics solution from raw data to executive presentation.

- ★ Created interactive Power BI dashboards with executive and customer-level views.
- ★ Developed executive KPI reporting with verified metrics.
- ★ Applied rigorous statistical validation (Pearson, T-Test, ANOVA, CI) to business findings.
- ★ Produced professional documentation, reports, and presentations.
- ★ Published a portfolio-ready GitHub repository.

Conclusion

The Sales Analytics End-to-End Project represents the successful completion of the ApexPlanet Data Analytics Internship and demonstrates the practical application of industry-standard analytics techniques.

From raw data preparation to executive storytelling, this project showcases the ability to solve real-world business problems through data-driven analysis, interactive visualization, statistical validation, and professional communication.

It serves as a portfolio-ready project that reflects both technical expertise and business understanding, preparing me for opportunities in **Data Analytics, Business Intelligence, and Data Science**.

GitHub Repository

github.com/shettyprasad-git/Sales-Analytics-End-to-End-Project-ApexPlanet

Prepared By

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Portfolio: prasad-shetty-portfolio.vercel.app

"Turning raw data into strategic business decisions through analytics, visualization, statistical validation, and executive storytelling."